SafeShare AI

On-Device AI Privacy Guardian for Screen Sharing

The Problem

• Screens often contain sensitive information
• Users may not notice it before sharing
• Cloud scanning creates an unnecessary privacy dependency
• Manual checking is slow and error-prone

Our Solution

• Scan a screenshot locally
• Detect sensitive text and visual elements
• Calculate a privacy-risk score
• Warn the user
• Blur, blackout or pixelate risky regions
• Export a privacy report

Real-World Example

Before sharing a screen:

Email + phone + account number + API key + QR code

SafeShare AI → detects → warns → protects

The user shares the protected version.

System Architecture

Screenshot
↓
Local preprocessing
↓
OCR + vision detectors
↓
Privacy rule engine
↓
Risk score
↓
Warning / protection
↓
Protected screenshot + CSV report

Snapdragon Connection

• Designed for Snapdragon-powered HP PCs
• Qualcomm AI Hub provides computer-vision and multimodal model categories
• AI Hub lists Snapdragon X Elite, X Plus 8-Core and X2 Elite targets
• Supported runtimes include ONNX Runtime and Qualcomm AI Runtime
• Final NPU claims will be based on actual device testing

Why On-Device AI?

Privacy: raw screen content can remain on the PC
Latency: no cloud round trip for core analysis

Offline operation: basic workflow does not need a cloud API
Deployment: model runtime can be optimized for target Qualcomm hardware

Core Features

■ PII pattern detection
■ Face detection
■ QR detection
■ Privacy-risk scoring
■■ Blur / blackout / pixelation
■ CSV privacy report

Innovation

The project turns privacy detection into an immediate protection workflow.

Detect → Explain → Protect

This makes the AI useful at the moment a user is about to expose their screen.

Technology

Python
Streamlit
OpenCV
Pillow
Pandas
OCR integration
ONNX Runtime deployment path
Qualcomm AI Hub optimization path

Validation Plan

• Test on the actual target HP Snapdragon laptop
• Verify the execution provider
• Measure latency and memory
• Test detection accuracy and false positives
• Capture screenshots of verified results
• Report only measured performance

Future Roadmap

Live screen guard
Custom privacy models
Better OCR
Enterprise privacy policies
Clipboard protection
Browser integration
Verified NPU acceleration

Closing

SafeShare AI
Your screen. Your data. Your privacy.